$$\tilde{\theta} = \frac{n_0 + n_2}{n}$$

$$\sup_{\theta \in \Theta} L(\theta) \longrightarrow \tilde{\theta} = \frac{n_0 + n_2}{n}$$

n_0 – число 0

n_2 – число 2

ОММП

b) $\theta = \bar{x}$ – несмещ.

$$\tilde{\theta} = \frac{n_0 + n_2}{n}$$

$$X_i = \begin{cases} 1, & x \in D \\ 0, & x \in C \end{cases}$$

$$\tilde{\theta} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

$$M X_i = 1 \cdot \theta + 0(1 - \theta) = \theta$$

$$M\tilde{\theta} = M \frac{1}{n} \sum X_i = \frac{1}{n} \sum M X_i = M X_i = \theta$$

несмещ.

c) $$I(\theta) = M\left[\left(\frac{\partial}{\partial \theta} \ln p\right)^2\right] =$$

$$= \int_{-1}^{1} \left(-\frac{1}{1-\theta}\right)^2 \cdot \frac{1-\theta}{2} dx + \frac{\theta}{2} \cdot \left(\frac{1}{\theta}\right)^2 + \frac{\theta}{2}\left(\frac{1}{\theta}\right)^2 =$$

$$= \int_{-1}^{1} \frac{1}{(1-\theta)2} dx + \frac{1}{2\theta} + \frac{1}{2\theta} =$$

$$= \frac{1}{1-\theta} + \frac{1}{\theta} = \frac{1}{\theta(1-\theta)} \; ; \; \theta \in (0;1)$$

~~$I(\theta)$ разрыв на Θ~~

~~и не положительна на Θ~~

~~$\Longrightarrow$ модель не регулярна $\Rightarrow$ не можем исп. неравенство Фишера~~

$$\left.\begin{array}{l} I(\theta) > 0 \\ I(\theta) - \text{непр.} \end{array}\right] \forall \theta \in (0;1)$$

$\Longrightarrow$ модель регулярна

$$D\tilde{\theta} = D\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum DX_i = \frac{1}{n^2} n \cdot \theta(1-\theta) =$$

$$= \frac{\theta(1-\theta)}{n} \text{ – ор. } \forall \text{ компакте } \boxed{\sim}$$

~~$D\theta$~~ для 1 события:

$$M\cancel{X}^2 = \int_{-1}^{3} x^2 \frac{1-\theta}{2} dx + 0^2 \cdot \frac{\theta}{2} + 2^2 \cdot \frac{\theta}{2} =$$

$$= \frac{2}{3} \cdot \frac{1-\theta}{2} + \cancel{\frac{1-\theta}{2}} + 2\theta = \frac{1-\theta}{3} + 2\theta$$

$$D\cancel{X} = \frac{1-\theta}{3} + 2\theta - \theta^2 = \frac{1}{3} + \frac{5\theta}{3} - \theta^2$$

Для θ:

$$D\theta = D\bar{X} = \frac{1}{n^2} \cdot n \cancel{\bar{X}} DX = \frac{1}{n} DX = \frac{1+5\theta-3\theta^2}{3n}$$

$$\frac{\theta(1-\theta)}{\cancel{n}} < \frac{1+5\theta-3\theta^2}{3\cancel{n}}$$

$$\theta - \theta^2 < 1 + 5\theta - 3\theta^2$$

$$0 \underset{\text{на } \boxed{\sim}}{<} 1 + 4\theta - 2\theta^2$$

$$\tilde{D\theta} = \frac{1}{n I(\theta)} = \frac{1}{n \frac{1}{\theta(1-\theta)}} = D\theta$$

$\Longrightarrow \tilde{\theta}$ – эфф. оценка в классе регулярных